# Double-Exponential Quasi-Orthogonality: The Geometry of Decoherence

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A composite system of $N$ local $q$-level factors has dimension $D = q^N$. Although at most $D$ vectors can be exactly orthogonal, a fixed squared-overlap tolerance $\varepsilon$ permits $M_\varepsilon(D) \gtrsim \exp(c_\varepsilon D)$ mutually quasi-orthogonal directions. Consequently $M_\varepsilon(q^N) \gtrsim \exp(c_\varepsilon q^N)$: the dimension grows exponentially in $N$, but its quasi-orthogonal capacity grows doubly exponentially. Lévy's lemma explains the accompanying concentration of regular observables, and Johnson–Lindenstrauss scaling gives the complementary finite-set account of exponential capacity. For Haar-random pure states the sharper exact law is $\mathbb{P}(|\langle\phi|\psi\rangle|^2 \geq \varepsilon) = (1-\varepsilon)^{D-1}$, while typical Fubini–Study angles lie within $O(D^{-1/2})$ of $\pi/2$: capacity explodes as angular structure homogenizes. We turn these facts into simultaneous overlap and trace-distance bounds for finite decoherence branch families, including mixed environments and collective weak coherences. The results are conditional on typical relative environmental dynamics. They neither select a pointer basis nor identify coherence suppression with readable or redundant records.

## I. INTRODUCTION

The quantum measurement problem includes several logically distinct questions: which observables acquire a preferred status, why outcome statistics obey the Born rule [1, 2] and why individual observations appear to have definite outcomes although the global state may remain a coherent superposition [3–5]. Seventeen years after introducing his famous cat thought experiment [6], Schrödinger returned to this tension in a 1952 Dublin colloquium. If the wave equation applies universally, he observed, a naive reading appears to turn macroscopic surroundings into a “quagmire” or “featureless jelly” and even seems to threaten the stability of an unobserved wristwatch in a drawer [7].

Environmental decoherence addresses an important operational part of this problem. It explains why interference between certain alternatives becomes inaccessible to measurements on a system once information about those alternatives has dispersed into its environment. Decoherence does not, by itself, select one member of the resulting global superposition as the unique outcome.

The modern operational response is not that the global superposition disappears. Rather, environmental correlations make its relative phases inaccessible to the restricted observables by which macroscopic systems are ordinarily interrogated.

Standard decoherence theory supplies the dynamical framework: the system–environment interaction selects stable pointer observables and correlates their alternatives with conditional environmental states [4, 8–10]. High dimension alone does not guarantee decoherence: a relative conditional unitary may, for example, remain the identity on the entire environmental space. The question addressed here is narrower: once an explicit typicality assumption for the conditional environmental states or relative unitaries has been specified, what suppression follows from high-dimensional geometry alone? Unit vectors sampled independently from a high-dimensional complex Hilbert space normally have very small overlaps. This is an elementary instance of concentration of measure [11–14].

The word *independently* is essential. Actual environmental branch states arise from the same initial state under correlated conditional dynamics. Their relative unitary need not be Haar distributed, and a large energy shell does not make it so. We therefore formulate the geometric results as conditional bounds, identify the precise dynamical object to which they would apply, and distinguish three notions that are easily conflated: small reduced-system coherences, distinguishable conditional environmental states, and redundant classical records.

The individual geometric identities used below are standard. The purpose of this paper is to combine them into operational trace-distance estimates for finite families of conditional environmental states, to extend the mean-square analysis to mixed environments, and to delineate the dynamical and information-theoretic assumptions needed before these kinematic estimates can be interpreted as physical decoherence claims. The organizing idea is that tensor composition and quasi-orthogonal packing are separate amplifications: the first makes dimension exponential in subsystem number, and the second makes the number of available almost-orthogonal directions exponential in that already enlarged dimension. Lévy concentration explains why this near-orthogonality is typical, while the Johnson–Lindenstrauss scaling explains, from the finite-set side, why the number of simultaneously representable directions is exponential in dimension.

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## II. TWO GEOMETRIC AMPLIFICATIONS

Consider $N$ local factors of dimension $q \geq 2$. Their finite tensor product has dimension

$$D = q^N. \tag{1}$$

This is the familiar exponential growth caused by composition. It is not, however, the only large-number effect. Exact orthogonality allows no more than $D$ mutually orthogonal vectors, but if the requirement is relaxed to

$$|\langle\psi_i|\psi_j\rangle|^2 \leq \varepsilon \qquad (i \neq j), \tag{2}$$

then projective Hilbert space admits families of cardinality $M_\varepsilon(D) \gtrsim \exp(c_\varepsilon D)$ for every fixed $0 < \varepsilon < 1$; the explicit random-coding bound is given in Eq. (31). Substitution of Eq. (1) gives

$$M_\varepsilon(q^N) \gtrsim \exp(c_\varepsilon q^N), \tag{3}$$

which is doubly exponential in the number of factors. The first exponential therefore comes from tensor composition, while the second comes from allowing almost, rather than exact, orthogonality.

The same phenomenon has a complementary angular description. For two independent Haar directions, the typical overlap amplitude is $O(D^{-1/2})$, so their Fubini–Study angle satisfies

$$\theta_{\mathrm{FS}} = \arccos|\langle\phi|\psi\rangle| = \frac{\pi}{2} - O(D^{-1/2}). \tag{4}$$

As $D$ grows, an enormous number of directions becomes available, yet most pairwise angles become nearly indistinguishable from a right angle. We call this *angular homogenization*, or a concentration-induced loss of angular resolution. It is not a literal loss of quantum information and it does not increase the linear dimension beyond $D$; it describes the coarsening of typical pairwise angular structure at fixed experimental resolution.

The two amplifications concern different variables. Tensor composition changes the ambient dimension as a function of $N$. Concentration and packing describe the distribution and number of directions inside that ambient space. The next two sections develop these statements through Lévy's and Johnson–Lindenstrauss lemmas before the exact overlap law is used to obtain the sharper operational estimates.

## III. LÉVY CONCENTRATION AND ANGULAR HOMOGENIZATION

Identify $\mathbb{C}^D$ with $\mathbb{R}^{2D}$ and write

$$\mathbb{S}^{2D-1} = \{|\psi\rangle \in \mathbb{C}^D : \langle\psi|\psi\rangle = 1\}. \tag{5}$$

Uniform measure on this sphere is the pure-state measure induced by Haar measure on $U(D)$. A real-valued function $f$ on the sphere is $L$-Lipschitz with respect to chordal distance if

$$|f(\psi) - f(\varphi)| \leq L\|\psi - \varphi\|. \tag{6}$$

In a convenient normalization, Lévy's lemma states that [13–15]

$$\mathbb{P}\{|f - \mathbb{E}f| \geq \delta\} \leq 2\exp\left[-\frac{(2D-1)\delta^2}{9\pi^3 L^2}\right]. \tag{7}$$

Versions centered at a median, and versions with different universal constants, express the same dimension-dependent concentration. The content is that every sufficiently regular scalar probe is almost constant over almost all of a high-dimensional sphere.

To connect this general theorem to quasi-orthogonality, fix a unit vector $|\phi\rangle$ and use

$$f_\phi(\psi) = |\langle\phi|\psi\rangle|^2. \tag{8}$$

Unitary invariance and normalization give

$$\mathbb{E}f_\phi = \frac{1}{D}. \tag{9}$$

Moreover,

$$\begin{aligned} |f_\phi(\psi) - f_\phi(\varphi)| &\leq \big(|\langle\phi|\psi\rangle| + |\langle\phi|\varphi\rangle|\big)|\langle\phi|\psi - \varphi\rangle| \\ &\leq 2\|\psi - \varphi\|, \end{aligned} \tag{10}$$

so $L \leq 2$. Equation (7) therefore yields

$$\mathbb{P}\left\{\left||\langle\phi|\psi\rangle|^2 - \frac{1}{D}\right| \geq \delta\right\} \leq 2\exp\left[-\frac{(2D-1)\delta^2}{36\pi^3}\right]. \tag{11}$$

Here concentration becomes quasi-orthogonality because the value about which the function concentrates is itself small. The sphere is overwhelmingly equatorial relative to any fixed direction: states with appreciable overlap occupy small spherical caps.

This application also shows exactly what the generic theorem does and does not provide. For a fixed excess threshold $\delta$, it gives an exponentially small exceptional measure. At the natural squared-overlap scale $\delta = O(D^{-1})$, however, the displayed generic bound becomes weak. The exact beta distribution in Lemma 1 resolves that scale and gives the sharper tail $\exp[-O(D\varepsilon)]$. Thus Lévy's lemma is the general geometric mechanism; the beta law is the specialized tool used for the quantitative decoherence and packing bounds below.

The corresponding projective statement is angular homogenization. Since the typical overlap amplitude is $O(D^{-1/2})$, the Fubini–Study angle obeys Eq. (4). Concentration therefore has two faces: almost all directions are almost orthogonal to a specified direction, and most angular comparisons become compressed into a narrow band about $\pi/2$. This is the precise geometric sense in which angular information becomes poorly resolved; no global state information is destroyed by the unitary theory.

## IV. JOHNSON–LINDENSTRAUSS SCALING AND FINITE CONFIGURATIONS

Lévy's lemma is a statement about measure on a sphere. The Johnson–Lindenstrauss lemma is complementary: it concerns the faithful representation of a specified finite metric configuration. For $0 < \delta < 1$ and a set of $M$ points in Euclidean space, there exists a map into $k$ dimensions, with

$$k \leq C\delta^{-2} \log M, \tag{12}$$

that preserves all squared pairwise distances within multiplicative factors $1 \pm \delta$ [16]. The universal constant $C$ and refinements of the dependence on $\delta$ are immaterial here; the decisive feature is the logarithmic dependence on $M$.

Apply this result to the $M$ standard basis vectors of $\mathbb{R}^M$ and normalize their projected images. Because their original norms are one and their mutual squared distances are two, preservation of norms and distances implies pairwise inner products of magnitude $O(\delta)$. Real vectors are a subset of complex Hilbert space, so the construction also supplies a complex spherical code. Reading Eq. (12) in the opposite direction gives

$$M \geq \exp(c\delta^2 D) \tag{13}$$

for a suitable fixed distortion and sufficiently large $D$. If the desired squared-overlap tolerance is denoted by $\varepsilon$, then $\varepsilon = O(\delta^2)$ and Eq. (13) has the same $\exp(c\varepsilon D)$ scale as the direct Haar random-coding bound.

The two results should nevertheless not be identified. Johnson–Lindenstrauss preserves the metric of an arbitrary finite point set after dimensional reduction; the packing argument asks how many normalized vectors can be placed in a fixed Hilbert space subject to an inner-product constraint. Using the standard basis connects them constructively, while the exact complex-Haar calculation below supplies the natural probability law and better constants for the present problem. Together, Lévy concentration and Johnson–Lindenstrauss scaling distinguish typicality from capacity: the former says that near-orthogonality occupies almost all directions, and the latter explains why an exponential number of finite directions can be accommodated at fixed resolution.

## V. DECOHERENCE FACTORS AND AN OPERATIONAL BOUND

Let $\{|s_i\rangle\}_{i=1}^m$ be orthonormal pointer alternatives spanning the support of the system state, selected by a measurement-like interaction, and let $\sum_i |c_i|^2 = 1$. For a pure initial environment $|E_0\rangle$, the joint evolution has the form $|E_i(t)\rangle = U_i(t)|E_0\rangle$, with

$$\left(\sum_{i=1}^m c_i|s_i\rangle\right)|E_0\rangle \longmapsto |\Psi(t)\rangle = \sum_{i=1}^m c_i|s_i\rangle|E_i(t)\rangle. \tag{14}$$

Tracing out the environment gives

$$\rho_S(t) = \sum_{i,j=1}^m c_i c_j^* \gamma_{ji}(t)|s_i\rangle\langle s_j|, \qquad \gamma_{ji}(t) = \langle E_j(t)|E_i(t)\rangle. \tag{15}$$

Let $\Delta$ denote complete dephasing in the pointer basis,

$$\Delta(\rho_S) = \sum_i |s_i\rangle\langle s_i|\rho_S|s_i\rangle\langle s_i|. \tag{16}$$

Thus $\Delta$ is trace preserving on the relevant support. All coherence statements below are relative to this dynamically selected pointer basis; they do not define a basis-independent quantity called coherence. The trace distance $D_{\rm tr}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ has a direct operational meaning: it controls the maximum change of outcome probabilities over all measurements. The environmental overlaps give the following elementary bound.

**Proposition 1** (Reduced-state coherence bound)**.** *For the state in Eq.* (15)*,*

$$D_{\rm tr}(\rho_S, \Delta(\rho_S)) \leq \min\left\{1, \frac{1}{2}\sum_{i\neq j} |c_i c_j|\, |\gamma_{ji}|\right\}. \tag{17}$$

*It also obeys the Hilbert–Schmidt bound*

$$D_{\rm tr}(\rho_S, \Delta(\rho_S)) \leq \min\left\{1, \frac{1}{2}\sqrt{mZ}\right\}, \tag{18}$$

*where*

$$Z = \sum_{i\neq j} |c_i|^2|c_j|^2|\gamma_{ji}|^2. \tag{19}$$

*If $|\gamma_{ji}| \leq \delta$ for all $i \neq j$, then*

$$\begin{aligned} D_{\rm tr}(\rho_S, \Delta(\rho_S)) &\leq \min\left\{1, \frac{\delta}{2}\left[\left(\sum_i |c_i|\right)^2 - 1\right]\right\} \\ &\leq \min\left\{1, \frac{m-1}{2}\delta\right\}. \end{aligned} \tag{20}$$

*For two alternatives the first inequality is an equality: $D_{\rm tr} = |c_1 c_2|\,|\gamma_{12}|$.*

*Proof.* Writing

$$X = \rho_S - \Delta(\rho_S) = \sum_{i\neq j} c_i c_j^* \gamma_{ji}|s_i\rangle\langle s_j|, \tag{21}$$

the triangle inequality, the trivial bound $D_{\rm tr} \leq 1$, and $\||s_i\rangle\langle s_j|\|_1 = 1$ give Eq. (17). Since the difference has rank at most $m$, $\|X\|_1 \leq \sqrt{m}\|X\|_2$ and direct evaluation of $\|X\|_2^2 = \mathrm{Tr}(X^\dagger X)$ give Eq. (18). The first bound in Eq. (20) follows by taking the maximum overlap out of the sum. The second follows from $\sum_i |c_i| \leq \sqrt{m}$ and $\sum_i |c_i|^2 = 1$. For $m = 2$, the two nonzero singular values of the off-diagonal difference are both $|c_1 c_2 \gamma_{12}|$. □

Pairwise quasi-orthogonality is therefore useful only together with control of the number and amplitudes of the alternatives. A tiny entrywise bound can fail to control a large Gram matrix if sufficiently many off-diagonal entries accumulate.

## VI. HAAR GEOMETRY AND SIMULTANEOUS BRANCH BOUNDS

### A. Exact overlap distribution

**Lemma 1** (Complex Haar overlap)**.** *Let $|\psi\rangle, |\phi\rangle \in \mathbb{C}^D$, $D \geq 2$, be independent Haar-random unit vectors. Then*

$$T = |\langle\phi|\psi\rangle|^2 \sim \mathrm{Beta}(1, D-1), \tag{22}$$

*and, for $0 \leq \varepsilon \leq 1$,*

$$\mathbb{P}(T \geq \varepsilon) = (1-\varepsilon)^{D-1} \leq e^{-(D-1)\varepsilon}. \tag{23}$$

*In particular, $\mathbb{E}T = 1/D$.*

*Proof.* By unitary invariance fix $|\phi\rangle = |e_1\rangle$. A Haar-random unit vector can be written as $Z/\|Z\|$, where the $Z_k$ are independent standard complex Gaussian variables. The variables $X_k = |Z_k|^2$ are independent exponentials of mean one. Hence

$$T = \frac{X_1}{X_1 + \cdots + X_D} \sim \mathrm{Beta}(1, D-1). \tag{24}$$

Integration of its density $(D-1)(1-t)^{D-2}$ gives Eq. (23); its mean is $1/D$. □

Thus the typical squared overlap is of order $D^{-1}$ and the typical amplitude is of order $D^{-1/2}$. The exact distribution is more informative than a generic application of Lévy's lemma because it retains the precise dependence on the overlap threshold. Equivalently, the Fubini–Study angle is typically $\pi/2 - O(D^{-1/2})$: concentration produces not only small overlaps but also the angular homogenization described in Sec. II.

### B. A finite-family theorem

The quantity relevant for a measurement with several alternatives is the largest pairwise overlap, rather than the overlap of one pair.

**Proposition 2** (Simultaneous overlap and trace-distance bound)**.** *Let $m \geq 2$, let $|E_1\rangle, \ldots, |E_m\rangle$ be independent Haar-random unit vectors in $\mathbb{C}^D$, and let $0 < \eta < 1$. Define*

$$K = \binom{m}{2}, \; x_\eta = \frac{\log(K/\eta)}{D-1}, \; \varepsilon_\eta = 1 - e^{-x_\eta}. \tag{25}$$

*With probability at least $1-\eta$,*

$$\max_{i\neq j} |\langle E_j|E_i\rangle| \leq \sqrt{\varepsilon_\eta} \leq \sqrt{\frac{\log[\binom{m}{2}/\eta]}{D-1}}. \tag{26}$$

*Consequently, the branched state in Eq. (14) obeys, with the same probability,*

$$\begin{aligned} D_{\mathrm{tr}}(\rho_S, \Delta(\rho_S)) &\leq \min\left\{1, \frac{1}{2}\left[\left(\sum_i |c_i|\right)^2 - 1\right]\sqrt{\varepsilon_\eta}\right\} \\ &\leq \min\left\{1, \frac{m-1}{2}\sqrt{\frac{\log[\binom{m}{2}/\eta]}{D-1}}\right\}. \end{aligned} \tag{27}$$

*Proof.* Lemma 1 and a union bound give

$$\mathbb{P}\left(\max_{i<j} |\langle E_j|E_i\rangle|^2 > \varepsilon\right) \leq \binom{m}{2}(1-\varepsilon)^{D-1}. \tag{28}$$

Equation (25) makes the right-hand side equal to $\eta$. The second inequality in Eq. (26) follows from $1 - e^{-x} \leq x$. Proposition 1 completes the proof. □

**Corollary 1** (Pure-Haar collective bound)**.** *Under the hypotheses of Proposition 2, with probability at least $1-\eta$,*

$$D_{\mathrm{tr}}(\rho_S, \Delta(\rho_S)) \leq \min\left\{1, \frac{1}{2}\sqrt{\frac{m}{D\eta}\left(1 - \sum_i |c_i|^4\right)}\right\}. \tag{29}$$

*Proof.* For $i \neq j$, Lemma 1 gives $\mathbb{E}|\gamma_{ji}|^2 = D^{-1}$. Therefore the quantity $Z$ in Eq. (19) satisfies

$$\mathbb{E}Z = \frac{1}{D}\left(1 - \sum_i |c_i|^4\right). \tag{30}$$

Markov's inequality gives $Z \leq \mathbb{E}Z/\eta$ with probability at least $1-\eta$. Substitution into Eq. (18) proves the claim. □

This collective estimate is complementary to Proposition 2: it can improve the branch-number scaling for broadly distributed weak coherences, whereas its confidence dependence is only $\eta^{-1/2}$ rather than logarithmic.

For fixed $m$ and fixed confidence, the operational error in this bound is $O(D^{-1/2})$. If $m$ grows with $D$, the accumulation factor in Eq. (27) must be retained. This qualification is important in discussions of very large branch families. For comparable amplitudes the displayed bound scales as $m\sqrt{\log(m)/D}$ and therefore becomes nontrivial in a considerably more restricted regime than the pairwise packing bound below. The union bound does not require mutual independence of the pair-overlap events. The same conclusion holds for any joint ensemble whose pairwise marginals have the Haar overlap distribution of Lemma 1.

The tradeoff can also be read directly from Eq. (25). At fixed $D$ and failure probability $\eta$, increasing the number of branches $m$ increases the overlap threshold $\varepsilon_\eta$ that can be guaranteed. Equivalently, keeping both $\eta$ and the allowed overlap fixed while increasing $m$ requires a larger effective dimension $D$.

### C. Packing is not classical memory capacity

The same union-bound calculation gives a random-coding statement. For $0 \le \varepsilon < 1$, let $M_\varepsilon(D)$ denote the largest cardinality of a family of unit vectors in $\mathbb{C}^D$ satisfying $|\langle\psi_i|\psi_j\rangle|^2 \le \varepsilon$ for every $i \ne j$. Then

$$M_\varepsilon(D) \ge \left\lfloor \exp\left[\frac{D-1}{2}\big(-\log(1-\varepsilon)\big)\right]\right\rfloor. \tag{31}$$

Indeed, for this number of independent Haar samples the union-bound probability of a violating pair is strictly below one. For fixed small $\varepsilon$, Eq. (31) is exponential in $\varepsilon D$. In particular, for $D = q^N$ it yields

$$M_\varepsilon(q^N) \ge \left\lfloor \exp\left[\frac{q^N-1}{2}\big(-\log(1-\varepsilon)\big)\right]\right\rfloor, \tag{32}$$

a doubly exponential growth with $N$. Likewise, for any fixed pair the failure probability in Eq. (23) is bounded by $\exp[-\varepsilon(q^N - 1)]$, which is doubly exponentially small in $N$. This second amplification is present only after the tolerance is fixed: the linear dimension remains $q^N$, and no more than $q^N$ states can be exactly orthogonal.

This direct complex-Haar construction realizes the same exponential scale as the Johnson–Lindenstrauss argument in Sec. IV. Here, however, the constant follows from the exact overlap tail, and the result directly controls the squared inner products relevant to environmental branches.

This spherical-code packing bound is not a capacity for reliably readable classical records. When $M > D$, the states are linearly dependent and cannot be perfectly discriminated by a single-copy measurement. The Welch bound further gives

$$\max_{i\ne j} |\langle\psi_i|\psi_j\rangle|^2 \ge \frac{M-D}{D(M-1)} \qquad (M > D), \tag{33}$$

Equation (33) is a geometric restriction, not by itself a bound on a specified decoder [17]. Approximate discrimination requires a prior distribution, a measurement, and an error criterion. For a uniform ensemble of $M$ pure states in dimension $D$, the Holevo bound $\chi \le \log D$, combined with Fano's inequality, gives a nonzero lower bound on the average decoding error when $\log M - \log D$ is sufficiently large [18, 19]. Equation (31) is therefore best viewed as a statement about projective Hilbert-space geometry. The operationally relevant decoherence statement is instead Proposition 2 for a specified family of alternatives.

## VII. MIXED ENVIRONMENTS: COHERENCE IS NOT A RECORD

This section isolates a counterintuitive but important distinction. Greater initial mixedness can reduce the Haar-averaged decoherence factor while simultaneously reducing, or even eliminating, the environment's capacity to encode a readable record. These are different operational quantities, not a paradox.

For an initially mixed environmental state $\rho_E$, a controlled unitary produces reduced-system coherence factors

$$\begin{aligned} \gamma_{ji}(t) &= \mathrm{Tr}\big[U_i(t)\rho_E U_j(t)^\dagger\big] = \mathrm{Tr}[\rho_E W_{ji}(t)]\,,\\ W_{ji}(t) &= U_j(t)^\dagger U_i(t). \end{aligned} \tag{34}$$

The relevant random object is the *relative* conditional unitary $W_{ji}(t)$, not two independently postulated purifications.

**Lemma 2** (Haar relative unitary). *Let $W$ be Haar-random on a $D$-dimensional Hilbert space and let $\rho$ be a fixed density operator, chosen independently of $W$, on that space. Then*

$$\mathbb{E}_W|\,\mathrm{Tr}(\rho W)|^2 = \frac{\mathrm{Tr}(\rho^2)}{D}. \tag{35}$$

*Proof.* In a fixed basis, Haar invariance gives the second-moment identity

$$\mathbb{E}_W[W_{ab}W^*_{cd}] = \frac{1}{D}\delta_{ac}\delta_{bd}. \tag{36}$$

Expanding $|\,\mathrm{Tr}(\rho W)|^2$ and applying this identity yields $D^{-1}\sum_{a,b}|\rho_{ab}|^2 = D^{-1}\,\mathrm{Tr}(\rho^2)$. □

The identity requires only the first unitary-design moment, equivalently the matrix-entry correlation $\mathbb{E}[W_{ab}W^*_{cd}]$. It immediately yields a simultaneous mixed-state statement.

**Proposition 3** (Mixed-environment simultaneous bounds). *Let $\mathcal{H}_D$ be a fixed $D$-dimensional environmental sector, let $\rho_E$ be a density operator on $\mathcal{H}_D$, and let the normalized coefficient vector $(c_i)$ be fixed independently of a specified ensemble of relative unitaries $W_{ji}$ acting on $\mathcal{H}_D$. Suppose that, for every $i < j$, the marginal ensemble of $W_{ji}$ reproduces the matrix-entry correlation $\mathbb{E}[W_{ab}W^*_{cd}] = D^{-1}\delta_{ac}\delta_{bd}$ on $\mathcal{H}_D$. The probability refers to the specified ensemble, such as Hamiltonian disorder, circuit randomness, conditional-perturbation randomness, or a prescribed random-time sampling procedure. No mutual independence of the different $W_{ji}$ is required. Put $K = \binom{m}{2}$ and let $0 < \eta < 1$. For every $\delta > 0$,*

$$\mathbb{P}\left\{\max_{i<j}|\gamma_{ji}| \ge \delta\right\} \le K\frac{\mathrm{Tr}(\rho_E^2)}{D\delta^2}, \tag{37}$$

*and hence, with probability at least $1-\eta$,*

$$\max_{i<j}|\gamma_{ji}| \le \min\left\{1, \sqrt{\frac{K\,\mathrm{Tr}(\rho_E^2)}{D\eta}}\right\}. \tag{38}$$

*Moreover, with probability at least $1-\eta$,*

$$D_{\mathrm{tr}}(\rho_S, \Delta(\rho_S)) \le \min\left\{1, \frac{1}{2}\sqrt{\frac{m\,\mathrm{Tr}(\rho_E^2)}{D\eta}\left(1 - \sum_i |c_i|^4\right)}\right\}. \tag{39}$$

*Proof.* Markov's inequality and Lemma 2 give $\mathbb{P}\{|\gamma_{ji}| \geq \delta\} \leq \mathrm{Tr}(\rho_E^2)/(D\delta^2)$. A union bound over the $K$ pairs proves Eqs. (37) and (38). For the stronger trace-distance estimate, define

$$Y = \sum_{i \neq j} |c_i|^2 |c_j|^2 |\gamma_{ji}|^2. \tag{40}$$

The assumed marginal second moments imply

$$\mathbb{E}Y = \frac{\mathrm{Tr}(\rho_E^2)}{D} \left(1 - \sum_i |c_i|^4\right). \tag{41}$$

Markov's inequality gives $Y \leq \mathbb{E}Y/\eta$ with probability at least $1-\eta$. Substitution into Eq. (18) proves Eq. (39). ☐

Writing $d_{\mathrm{purity}} = 1/\mathrm{Tr}(\rho_E^2)$ makes the mean-square scale

$$\frac{\mathrm{Tr}(\rho_E^2)}{D} = \frac{1}{D d_{\mathrm{purity}}}. \tag{42}$$

Thus $D$ characterizes the relative-unitary ensemble, whereas $d_{\mathrm{purity}}$ characterizes the initial environmental mixedness.

For a pure environment, Eq. (35) reduces to $1/D$. For the maximally mixed state it gives $1/D^2$. This stronger suppression does *not* mean that a maximally mixed environment stores a better record. The conditional environmental states are

$$\rho_E^{(i)} = U_i \rho_E U_i^\dagger. \tag{43}$$

If $\rho_E = I/D$, all of them are identical, so no measurement on the environment can reveal $i$, even though $\mathrm{Tr}(W_{ji})/D$ may be small.

For pure conditional states, the connection between overlap and record distinguishability is direct. With equal priors, the minimum error probability for discriminating two pure states is

$$p_{\mathrm{err}} = \frac{1}{2}\left(1 - \sqrt{1 - |\langle E_j | E_i \rangle|^2}\right). \tag{44}$$

Equivalently,

$$D_{\mathrm{tr}}(|E_i\rangle\langle E_i|, |E_j\rangle\langle E_j|) = \sqrt{1 - |\langle E_j | E_i \rangle|^2}. \tag{45}$$

For mixed states, however, distinguishability is governed by trace distance or fidelity between the density operators in Eq. (43), not by a chosen purification overlap alone. In particular, unitary invariance gives

$$D_{\mathrm{tr}}\left(\rho_E^{(i)}, \rho_E^{(j)}\right) = \frac{1}{2}\left\| W_{ji} \rho_E W_{ji}^\dagger - \rho_E \right\|_1, \tag{46}$$

which is generally unrelated to the magnitude of $\mathrm{Tr}(\rho_E W_{ji})$. Decoherence, locally accessible records, and redundant records are therefore different properties. The latter requires information about the pointer variable to be recoverable from many disjoint environmental fragments [20–22]. Indeed, generic global random states need not have the branching structure needed for redundant storage [20]. For initially mixed environments or more general mixed global states, system decoherence can occur without system–environment entanglement or objective records [23]. By contrast, for an initially pure global state, nontrivial reduced-state mixedness entails system–environment entanglement.

## VIII. THE DYNAMICAL BRIDGE

The preceding propositions are kinematic. In a microscopic pure-dephasing model with time-independent conditional environmental Hamiltonians $H_i$,

$$\gamma_{ji}(t) = \langle E_0 | e^{iH_j t} e^{-iH_i t} | E_0 \rangle. \tag{47}$$

Its modulus squared is a Loschmidt echo. Decoherence can consequently be studied using fidelity decay, spectral correlations, perturbation theory, and many-body orthogonality effects [24, 25]. Equation (47) also exposes why a large Hilbert space is insufficient: $e^{iH_j t} e^{-iH_i t}$ may remain close to the identity, preserve large invariant subspaces, or display revivals long before any Poincaré recurrence.

Several dimensions should be distinguished. For an energy window $[E, E + \Delta E]$, let

$$D_{\mathrm{shell}} = \mathrm{Tr}\ \mathbf{1}_{[E,E+\Delta E]}(H_E). \tag{48}$$

The Boltzmann entropy associated with that convention is

$$S_B(E, \Delta E) = k_B \log D_{\mathrm{shell}}. \tag{49}$$

By contrast, let $H_E = \sum_\lambda E_\lambda P_\lambda$ be the spectral decomposition into distinct energy eigenspaces and put

$$p_\lambda = \langle E_0 | P_\lambda | E_0 \rangle, \qquad d_{\mathrm{spec}} = \frac{1}{\sum_\lambda p_\lambda^2}. \tag{50}$$

This spectral participation ratio measures how broadly the initial state is distributed over distinct energy eigenspaces and is widely used in equilibration bounds [26–28]. For a nondegenerate Hamiltonian and $|E_0\rangle = \sum_\alpha a_\alpha |E_\alpha\rangle$, it reduces to $d_{\mathrm{spec}} = 1/\sum_\alpha |a_\alpha|^4$. The projector formulation avoids dependence on an arbitrary basis chosen inside a degenerate eigenspace. Neither $D_{\mathrm{shell}}$ nor $d_{\mathrm{spec}}$ is automatically the dimension $D$ appearing in a Haar model for $W_{ji}(t)$.

For chaotic many-body systems, eigenstate thermalization and spectral statistics offer reasons to expect equilibration of appropriate observables, but they do not imply that a single finite-time orbit is Haar-random over an entire shell [29, 30]. Conserved charges, locality, finite propagation speed, integrability, many-body localization, and weak conditional perturbations may all obstruct the Haar model. A defensible application therefore requires a statement about the ensemble or time distribution of the relative unitaries $W_{ji}(t)$. At a fixed time, fixed Hamiltonians and a fixed initial state determine $W_{ji}(t)$ deterministically. Every probability statement must therefore identify its probability source, such as Hamiltonian disorder, a random circuit ensemble, random initial states, a distribution of conditional perturbations, or a random time sampled from a specified window. A temporal distribution along one orbit agrees with an ensemble distribution only under an additional ergodicity or sampling assumption and generally contains correlated samples. For time-dependent $H_i(t)$, the exponentials in Eq. (47) must be replaced by the appropriate time-ordered propagators.

Approximate unitary designs provide one possible intermediate assumption. The mean-square identity in Lemma 2 requires only the relevant second moment, whereas a tail comparable to Eq. (23) requires stronger moment or concentration control. Local random circuits are known to approach approximate designs only after a depth that depends on locality, system size, design order, and accuracy [31]. This makes circuit depth or physical evolution time a meaningful replacement for an unqualified appeal to Haar typicality.

## IX. A LOCAL PRODUCT MECHANISM

Many decoherence models have a more physical route to small overlaps than global Haar randomness. As an idealized mechanism, suppose that the environment is initially factorized, residual interactions between fragments can be neglected on the timescale of interest, and the conditional dynamics preserves the product form

$$|E_i\rangle = \bigotimes_{k=1}^{N} |e_i^{(k)}\rangle. \tag{51}$$

Consequently,

$$\begin{aligned} \langle E_j|E_i\rangle &= \prod_{k=1}^{N} \langle e_j^{(k)}|e_i^{(k)}\rangle, \\ -\log|\langle E_j|E_i\rangle| &= \sum_{k=1}^{N} -\log|\langle e_j^{(k)}|e_i^{(k)}\rangle|. \end{aligned} \tag{52}$$

The logarithmic identity assumes nonzero local overlaps; if one local overlap vanishes, the global overlap vanishes already. If a positive density of fragments has local overlaps bounded above by a fixed $q < 1$, or more generally if

$$\liminf_{N\to\infty} \frac{1}{N} \sum_{k=1}^{N} \left[-\log|\langle e_j^{(k)}|e_i^{(k)}\rangle|\right] > 0, \tag{53}$$

Eq. (52) gives exponential suppression in $N$ without requiring the global state to be Haar-random. At the same time, the fragment structure makes the question of redundant records well posed. This local mechanism and the global geometric estimate are complementary, not interchangeable. Approximate rather than exact conditional factorization produces model-dependent correction terms.

## X. THERMODYNAMIC-LIMIT COMPARISON: INFINITE TENSOR PRODUCTS AND SECTORIZATION

The finite-dimensional estimates above concern generic Haar-distributed pairs, whose overlaps are nonzero almost surely but typically very small. Finite-dimensional Hilbert spaces can of course contain exactly orthogonal states, and a finite product overlap is exactly zero if even one corresponding pair of local factors is orthogonal. A related but conceptually distinct phenomenon occurs when macroscopically different reference sequences define disjoint representation sectors in an infinite tensor-product limit.

Here $N$ denotes the number of tensor factors, whereas $D$ in the preceding sections denotes the dimension of the sector on which the random-state or relative-unitary model is imposed; in many-body applications $D$ may itself grow exponentially with $N$. Consider two sequences of normalized local states,

$$|E_i^{(N)}\rangle = \bigotimes_{k=1}^{N} |e_i^{(k)}\rangle, \qquad |E_j^{(N)}\rangle = \bigotimes_{k=1}^{N} |e_j^{(k)}\rangle. \tag{54}$$

Their overlap factorizes as

$$\langle E_j^{(N)}|E_i^{(N)}\rangle = \prod_{k=1}^{N} \langle e_j^{(k)}|e_i^{(k)}\rangle. \tag{55}$$

Let $n_N$ be the number of factors among the first $N$ for which

$$\left|\langle e_j^{(k)}|e_i^{(k)}\rangle\right| \le q < 1, \tag{56}$$

and suppose that $n_N/N \to \alpha > 0$. Then the overlap vanishes exponentially,

$$\begin{aligned} \left|\langle E_j^{(N)}|E_i^{(N)}\rangle\right| &\le q^{n_N} = \exp[-n_N|\log q|] \\ &= \exp[-\alpha N|\log q| + o(N)] \longrightarrow 0, \end{aligned} \tag{57}$$

Thus the sequence of finite-product overlaps tends to zero in the thermodynamic limit. This limiting statement alone does not yet establish inequivalent representations: an infinite tensor-product representation and an algebra of quasi-local observables must also be specified.

Von Neumann's construction of infinite tensor products makes this limit more precise [32]. For two normalized reference sequences, a standard sufficient criterion for their failure to belong to the same weak equivalence class is

$$\sum_{k=1}^{\infty} \left(1 - |\langle e_j^{(k)}|e_i^{(k)}\rangle|\right) = \infty. \tag{58}$$

The positive-density condition above implies this divergence. In the standard infinite-product construction, such reference sequences can define orthogonal incomplete tensor-product sectors and, under the corresponding quasi-local formulation, disjoint or unitarily inequivalent representations of the observable algebra. Here "unitarily inequivalent" concerns representations of the physical observable algebra, not the absence of an abstract isomorphism between Hilbert spaces of the same dimension.

This distinction is important. At every finite $N$, all product states are vectors in the same finite-dimensional Hilbert space and can be related by finite-system unitaries. Different thermodynamic-limit states may instead define disjoint representations of the quasi-local algebra. The latter conclusion is representation dependent and does not follow from finite-$N$ geometry alone. It is one mathematical framework used in thermodynamic-limit approaches to measurement theory [33–35].

For infinitely many nontrivial finite-dimensional factors, the complete infinite tensor product is generally nonseparable and decomposes into mutually orthogonal sectors, while an individual incomplete tensor-product sector can remain separable. The complete product may carry continuum many sector labels. Writing this cardinal specifically as $\aleph_1$ assumes the continuum hypothesis. The physically relevant distinction is the emergence of disjoint representations of the quasi-local observable algebra, not the cardinality label by itself.

This infinite-system mechanism may therefore be viewed as an idealized exact counterpart of the finite-dimensional mechanisms discussed above: high-dimensional Haar sectors and finite product environments satisfying Eq. (53) can yield parametrically small overlaps, with exponential suppression in $N$ in the product case, whereas a specified infinite-product construction can yield disjoint sectors. It does not, by itself, establish that a physical environment is literally infinite, select one sector as the unique realized outcome, or derive the Born probabilities. It does permit macroscopically distinct phases or outcomes to be represented in inequivalent sectors rather than forcing them into a single irreducible representation.

## XI. SCOPE OF THE GEOMETRIC STATEMENT

The geometric calculation establishes the following conditional implication: if a specified finite family of environmental branches is distributed like independent typical vectors in a $D$-dimensional sector, then its largest pairwise overlap and the operational distance of the reduced system from its dephased state are small with high probability. It also shows that the projective Hilbert space contains exponentially large pairwise quasi-orthogonal codes. When $D = q^N$, composition first creates an exponentially large ambient dimension and quasi-orthogonal packing then creates a doubly exponential number of directions at fixed overlap tolerance. Simultaneously, concentration compresses most pairwise angular differences into a narrow neighborhood of $\pi/2$.

It does not establish any of the following claims. First, it does not select a pointer basis; this depends on the structure and timescales of the system–environment interaction. Second, it does not prove that a physical Hamiltonian generates Haar-typical relative branches. Third, it does not turn an improper mixture into a proper ensemble or select a unique outcome. Fourth, it does not equate small system coherence with a readable, let alone redundant, environmental record. Finally, exponential spherical packing does not provide exponentially many perfectly decodable classical labels.

The operational content relevant to Schrödinger's "jellification" worry is therefore limited but precise. For every POVM effect $0 \le M \le I$,

$$\left|\mathrm{Tr}\big[M\big(\rho_S - \Delta(\rho_S)\big)\big]\right| \le D_{\mathrm{tr}}(\rho_S, \Delta(\rho_S))\,. \tag{59}$$

The supremum of the left-hand side over all effects equals the trace distance. For a Hermitian observable $A$ with $\|A\|_\infty \le 1$, the corresponding expectation-value difference is bounded by $2D_{\mathrm{tr}}$. Propositions 2 and 3 therefore directly bound changes in system measurement probabilities under their respective typicality assumptions. They do not assert that the global coherent superposition has vanished.

## XII. FURTHER RESEARCH

A first concrete test is to replace the Haar assumption by conditional local Hamiltonians $H_i = H_E + V_i$. One can compare chaotic, integrable, and localized spin environments by measuring the decay, long-time plateau, temporal variance, and full distribution of Eq. (47). Histograms of $|\gamma_{ij}(t)|^2$ can be compared directly with the $\mathrm{Beta}(1, D-1)$ law in Lemma 1, but the sampling prescription must be stated. An ensemble histogram and a time histogram along one correlated orbit are not interchangeable without an ergodicity and sampling analysis. Their deviation from the beta law is itself a measure of the failure of pair typicality.

A second problem is finite-time design formation. For conditional random circuits or noisy Floquet systems, one can seek explicit bounds of the form

$$\mathbb{P}\{|\,\mathrm{Tr}(\rho_E W_{ji}(t))|^2 \ge \varepsilon\} \le f(D, t, \varepsilon, \mathrm{Tr}\,\rho_E^2), \tag{60}$$

interpolating between short-time perturbative behavior and a random-matrix plateau. Here $\mathbb{P}$ must refer to a specified circuit, disorder, initial-state, perturbation, or random-time ensemble. Locality should appear explicitly through circuit depth or a Lieb–Robinson scale.

A third direction is the many-branch problem. Rather than controlling only the largest entry, one can study the spectrum of the Gram matrix $G_{ij} = \langle E_j | E_i \rangle$ and derive bounds tailored to the coefficient vector $(c_i)$. This would sharpen Eq. (17) and identify when numerous weak coherences collectively remain observable.

A related geometric problem is to replace Haar measure by physically generated ensembles while retaining a concentration principle. One may ask whether the relevant conditional circuit or Hamiltonian ensemble satisfies a logarithmic Sobolev inequality, a spectral-gap estimate, or an approximate design condition strong enough to reproduce a Lévy-type tail for selected Lipschitz observables. On the finite-family side, one can compare the Gram matrices of dynamically generated branches with Johnson–Lindenstrauss-type near-isometries. Such results would connect the general concentration and embedding principles to realistic, non-Haar dynamics rather than merely postulating pair typicality.

Finally, for mixed environments one should analyze two quantities in parallel: the decoherence factor $\mathrm{Tr}(\rho_E W_{ji})$ and the distinguishability or accessible information of the conditional fragment states $\rho_F^{(i)}$. Their separation is necessary for determining when geometric decoherence is accompanied by objective, redundantly accessible records.

## XIII. CONCLUSION

The central geometric effect is a combination of abundance and featurelessness. Tensor composition gives $D = q^N$, exponentially many linear dimensions in the number of factors. Relaxing exact orthogonality to a fixed overlap tolerance then permits $\exp(c_\varepsilon D)$ directions, hence a doubly exponential number in $N$. Yet typical overlaps are only $O(D^{-1/2})$ in amplitude, so almost all projective angles lie within $O(D^{-1/2})$ of $\pi/2$. Dimensional capacity therefore explodes while typical angular structure becomes increasingly homogeneous.

Lévy's lemma supplies the general reason: regular observables concentrate on the high-dimensional unit sphere, and the overlap observable concentrates about the small mean $1/D$. The exact beta law sharpens this general result at the natural overlap scale. Johnson–Lindenstrauss supplies the complementary finite-configuration statement: logarithmic dimension suffices to retain the pairwise geometry of exponentially many points. The direct Haar packing calculation joins these two perspectives for complex projective Hilbert space.

This geometry supplies a useful but conditional component of decoherence. Independent Haar-random pure states have the exact overlap tail $(1-\varepsilon)^{D-1}$. For a finite family, this tail yields a high-probability bound on both the largest branch overlap and the trace distance between the reduced system and its pointer-basis dephasing. For a mixed initial environment, Haar-random relative dynamics instead gives the mean-square coherence $\mathrm{Tr}(\rho_E^2)/D$ and the simultaneous bounds of Proposition 3. The Hilbert–Schmidt estimate (18) additionally controls the collective contribution of many weak coherences. It provides a complementary collective bound and can be substantially sharper than the entrywise estimate when many weak coherences are broadly distributed, although it is not uniformly sharper.

The packing number is not an additional Hilbert-space dimension, and angular homogenization is not literal destruction of global information. These results quantify geometric capacity, not dynamics. The physical question is whether conditional relative unitaries generated by a particular local Hamiltonian acquire the necessary typicality on the relevant timescale. Moreover, loss of system coherence, environmental distinguishability, and redundant record formation remain separate notions. With these distinctions made explicit, high-dimensional geometry provides a clean quantitative bound on locally accessible interference without being asked to solve the preferred basis, definite-outcome, or objectivity problems by itself.

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